

SANJEEV KUMAR

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EDUCATION

Delhi Technological University (Formerly DCE)

B.Tech in Computer Science & Engineering, Minor in Machine Learning

New Delhi, India

Nov 2022 – Jun 2026

Yuva Shakti Model School

Delhi, India

Class XII (CBSE) | **90.8%**

2022

Yuva Shakti Model School

Delhi, India

Class X (CBSE) | **93.2%**

2020

EXPERIENCE

Outlier AI

Remote

AI Training Specialist & QA Reviewer

Jan 2026 – Apr 2026

- Evaluated and quality-reviewed **500+ LLM-generated responses** across coding, mathematics, and statistical reasoning for correctness, instruction adherence, and reasoning quality using structured evaluation rubrics.
- Performed systematic **error analysis** on model failures and edge cases, identifying recurring logical, factual, and instruction-following issues to improve evaluation consistency and training-data quality.
- Documented failure patterns, edge cases, and quality findings for downstream review, supporting more reliable **LLM evaluation and QA workflows**.

The Sparks Foundation

Remote

Data Science & Business Analytics Intern

Feb 2025 – Mar 2025

- Performed **exploratory data analysis** using Python, Pandas, and NumPy to identify inventory trends, low-turnover product categories, and operational inefficiencies.
- Cleaned and analyzed inventory data, handling missing values and outliers to evaluate product performance and identify categories contributing to higher inventory holding costs.
- Built interactive **Tableau dashboards** to track business KPIs, inventory trends, and product-level performance, enabling clearer data-driven analysis of operational metrics.

PROJECTS

GRACE - Efficient Multimodal Hateful Meme Detection | PyTorch, CLIP, LLaVA, Cross-Attention

[GitHub](#)

- Contributed to a multimodal classification framework using **PyTorch, CLIP-based image/text representations, asymmetric dual-path cross-attention, and offline LLM-rationale distillation** for efficient image-text reasoning.
- Evaluated the system across Hateful Memes Challenge, HarMeme, and PrideMM, achieving **83.51%, 93.14%, and 85.72% AUROC**, respectively.
- Performed **latency benchmarking, ablation studies, hyperparameter analysis, and failure analysis**; the final system achieved **21 ms inference per meme on a single NVIDIA T4 GPU**.

SochSamajh AI – Intelligent Query Routing System | Python, FastAPI, LangGraph, LangSmith, Docker

[GitHub](#)

- Engineered a modular query-routing backend with LangGraph and FastAPI, achieving **83% routing accuracy** and **79.7% macro-F1** across a **300-query benchmark** using heuristic classification with an LLM fallback for ambiguous requests.
- Designed independent safety, routing, critic, and response-formatting components, exposed through **FastAPI REST endpoints** for maintainable service integration.
- Used **LangSmith tracing** to debug execution paths, identify failure cases, and improve system reliability across routing decisions.

Hybrid Document Search & QA System | Python, FAISS, BM25, LangChain

[GitHub](#)

- Built a hybrid **retrieval-augmented generation pipeline** combining FAISS dense retrieval with BM25 sparse retrieval for document-grounded question answering.
- Benchmarked retrieval across **10+ document sets** and reduced hallucination rate by **30%** versus naive prompting through experiments with chunking, retrieval, and context selection.

TECHNICAL SKILLS

Languages: C++, Python, Java, TypeScript, JavaScript, SQL

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, MySQL, MongoDB, Redis, Prisma, REST APIs

AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, LangChain, LangGraph, RAG, ChromaDB, BM25

Tools & Platforms: Git, Docker, AWS, LangSmith, Streamlit, GitHub Actions, n8n

ACHIEVEMENTS

Competitive Programming: Solved **800+** DSA problems across LeetCode and Codeforces; LeetCode Max Rating **1716** (Top 12%), Codeforces Specialist (**1517**).

Academic: Scored **98.87 percentile** in JEE Main.

Hackathons: Participated in **SignalWatch** and the on-campus **HCL Hackathon**, gaining hands-on experience in collaborative problem solving and rapid prototyping.

Leadership & Volunteering: Volunteered at **Engifest**, DTU's annual cultural festival, supporting event coordination and operations.

Certifications: Andrew Ng's Machine Learning & Deep Learning Specializations (DeepLearning.AI).